MODEL QUESTION PAPER

TED(15) – 5094 Reg. No……………

(REVISION – 2015) Signature…………….

FIFTH SEMESTER DIPLOMA EXAMINATION IN POLYMER TECHNOLOGY

INJECTION MOULD DESIGN

[*Time* : 3 hours

(Maximum marks : 100)

PART – A

(Maximum marks : 10)

I Answer all questions in one or two sentences. Each question carries 2 marks Marks

1. State the purpose of lathe.
2. Differentiate cavity and core of a mould.
3. Write any two advantages of ejector plate ejection over pin ejection system
4. What do you mean by parting line.
5. Define facing operation in a lathe. (5 x 2 = 10)

PART – B

(Maximum marks : 30)

II. Answer *any five* of the following questions. Each question carries 6 marks.

1. Draw neat sketch of a shaping machine and label its parts.
2. Differentiate sprue and sprue bush in a mould.
3. What are the advantages of air ejection method.
4. Explain the method of cooling integer type core plate
5. Describe the function of register ring and guide pillar in an injection mould.
6. State the design consideration for positioning of gate in a mould assembly
7. Illustrate frame type bolster plate (5 x 6 = 30)

PART – C

(Maximum marks : 60)

(Answer one full question from each module. Each full question carries 15 marks)

Module – I

III (a). With a neat sketch explain the parts of a horizontal type milling machine. (8)

(b) Explain the sequence of operations for making a guide pillar of a mould in alathe. (7)

OR

IV (a) Draw a neat sketch of a shaping machine and explain functions of each parts. (8)

(b) State the difference between planer and shaper. (7)

Module –II

V (a) Explain pressure casting method for making cavity insert. (8)

(b) Describe the methods of fitting inserts into a bolster plate. (7)

OR

VI (a) Explain cold hobbing process for making cavity inset. (8)

(b) List the seven stages involved in bench fitting. (7)

Module –III

VII (a) Explain stripper bar ejection techniques. (8)

(b) What is the function of ejector grid ? Describe inline ejector grid. (7)

OR

VIII (a) Explain air ejection techniques. (8)

(b) Describe ejector plate return system by spring return method. (7)

Module –IV

IX (a) Explain profiled parting surface and angled parting surface . (8)

(b) Describe the cooling system for integer type core plate of a mould . (8)

OR

X (a) Explain sprue gate and rectangular gate (8)

(b) What are the methods generally adopted for mould cooling. (7)

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